BATTERY MANAGEMENT SYSTEM FOR EV APPLICATION

A PROJECT REPORT

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by

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CERTIFICATE

This is to certify that the project work titled **"BATTERY MANAGEMENT SYSTEM FOR EV APPLICATION"** submitted by ***SARAVANAN S*** , ***RANGANATHAN V*** , ***PRAVEEN KUMAR P*** is in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY**, is a record of bona fide work done under my guidance. The contents of this project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for award of any degree or diploma and the same is certified.

8/4/2026

Dr.Krishna Kumba

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The thesis is satisfactory / u~~nsatisfac~~tory

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The student is free to acknowledge all those he feels he should acknowledge on the basis of the guidance and help provided during the implementation of the project. If the student has conducted his project elsewhere (viz. outside VIT) appropriate acknowledgement should be given to all concerned.

It is customary to acknowledge the University Management / respective School Director for giving the candidate an opportunity to carry out his /her studies at the University.

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ABSTRACT

This project will involve the designing and creation of an intelligent Battery Management System (BMS) through the use of an STM32F407VG Discovery microcontroller for effective battery management. Several sensors will be used in order to achieve this goal effectively. The system is essential since it plays a role in ensuring that batteries are monitored and controlled effectively during operations in order to avoid issues such as overheating which could damage or reduce the efficiency of the batteries. Voltage monitoring will involve the use of the INA219 with STM32 chip . Current measurement will be achieved through the use of the INA219 sensor. Thermal control will be achieved by measuring the battery temperatures using the LM35 sensor.The measured values will include voltage, current, power consumption, and state of charge. They will be displayed on a 16x2 LCD to allow easy identification of battery status. The system also involves a DC motor load OF 12V,1000rpm . In order to enhance its functionality, the ESP32 microcontroller will be added to the system to monitor batteries based on the Internet of Things (IoT). This allows transmission of the data to the Blynk IoT application.It gives users the ability to keep track of battery status and get immediate alerts on their mobile phones in case there is an anomaly like overvoltage, overcurrent, and over-temperature. Another protective measure introduced by the system involves using a relay circuit that will switch off the load when unsafe conditions prevail. In summary, this project has developed an efficient, affordable, and intelligent BMS system that can be used in modern-day battery applications

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ABBREVIATIONS AND NOMENCLATURE

UART	Universal Asynchronous Receiver/Transmitter
I2C	Inter-Integrated Circuit
BMS	Battery Management System
IoT	Internet of Things
EV	Electric Vehicle
LCD	Liquid Crystal Display
SoC	State Of Charge
SoH	State Of Health
ADC	Analog to Digital Converter
MCU	Microcontroller Unit
GPIO	General Purpose Input/Output
PCB	Printed Circuit Board

CHAPTER I

1. INTRODUCTION

1.1 INTRODUCTION

The rising trend of electric vehicle (EV) usage has underlined the importance of implementing efficient Battery Management Systems (BMS). Batteries such as Lithium-ion batteries that are used in EVs can be affected by parameters like voltage, current, and temperature. As a result, continuous monitoring of these factors is vital to avoiding dangerous scenarios that include overcharge, deep discharge, and even thermal runaway. In this project, we will focus on creating an embedded BMS with STM32F407VG Discovery microcontroller. Our system will monitor essential factors for a 3S battery pack using specific sensors such as INA219 for measuring current and LM35 for measuring the temperature. Voltage will be detected with the help of the STM32 microcontroller through INA219. Collected information will be displayed locally with the use of the LCD screen.Further, the SoC Estimation feature has been used for estimating the state of charge to determine the battery status, thereby improving the ease-of-use in electric vehicle applications. Furthermore, information about our battery pack will be sent to the Blynk IoT platform using ESP32. Finally, to make our test more realistic, we will create an additional load with a motor of 12V,1000rpm .Protection strategies like overvoltage, overcurrent, and overtemperature protection have been integrated by the use of the relay module. This strategy is efficient and economical since it can be scaled to accommodate future changes in EV batteries.

1.1.1 Motivation

The rationale for undertaking this study emanates from the need for the safe and efficient management of the energy consumption and storage in modern day electrical vehicles. Inaccurate monitoring of the battery parameters may not only cause inefficiency in battery use but also poses hazards to the consumers and their environment. The use of commercially available battery management systems may be expensive and cumbersome, and thus they are less applicable in educational purposes. The objective of this project is to build a cost-effective and efficient battery management system.The use of the IoT system in embedded systems helps improve user awareness, increase battery life, and ensure the safety of electric vehicles.

1.1.2 Objectives

The main objectives of this project are:

- To design a Battery Management System using the STM32F407VG microcontroller for EV applications.
- To determine battery voltage using INA219.
- To monitor and control current using the INA219 current sensor.
- To measure temperature using the LM35 sensor for thermal protection.
- To display real-time parameters (voltage, current, temperature) on an LCD.
- To simulate load conditions using a motor of 12V, 1000rpm
- To estimate and display the State of Charge (SoC) of the battery.
- To implement protection mechanisms (overvoltage, overcurrent, overtemperature) using a relay module.
- To integrate an ESP32 microcontroller for IoT-based data transmission.
- To send real-time battery data and alerts to the Blynk IoT application for remote monitoring.

1.1.3 Scope of the Work

The extent of work that will be carried out in the development of the BMS includes its design, development, and installation to provide real-time monitoring, control, and protection of the battery pack in the EVs. These tasks include:

- Monitoring of voltage, current, and temperature using sensors and estimation SoC
- LCD display of parameters.
- Load simulation using a motor and driver module.
- Wireless transmission of data via ESP32 to Blynk IoT platform.
- Alert and notifications in case of any problem.
- Protection via relays.

This work does not involve battery balancing or professional safety equipment but only a prototype-level implementation of the proposed project. However, there are possible developments in the future to integrate SoH prediction algorithms using machine learning techniques, cloud computing to store the generated data, and further enhancement on cell balancing, among others.

1.2 ORGANIZATION OF THESIS

The thesis will be divided into five chapters, each covering a particular stage of the project.

Chapter 1: Introduction:

Gives an introduction to the project and its motive, objective, and scope concerning designing a smart Battery Management System for EV applications.

Chapter 2: Project Description:

Discusses the working mechanism of the system involving various modules like sensing, internet of things, protection, and load simulation.

Chapter 3: Project Design:

Discusses the design aspect of the project involving the interface of STM32 and other sensors, circuit designing, and communication protocol designing.

Chapter 4: Project Demonstration:

Includes details of implementing the project, testing procedure, test results, and their analysis.

Chapter 5: Conclusion and Future Scope:

Will include the findings of the project as well as future scope related to using analytics, cloud, and machine learning-based intelligent battery management systems.

CHAPTER II

2. PROJECT DESCRIPTION

2.1 OVERVIEW OF PROJECT

This project involves designing and implementation of the BMS (Battery Management System) with regard to EVs (Electric Vehicles). The project targets development of the system able to continuously control critical parameters of the 3S lithium-ion battery pack including the battery voltage, current, temperature, and SoC (State of Charge). In order to measure voltage, INA219 will be used, as well the INA219 will allow for measuring current and the LM35 sensor – for temperature measurement. All these parameters are to be presented in real time on the LCD. In order to simulate real conditions for EV usage. Moreover, an ESP32 microcontroller provides the possibility to monitor the battery parameters in terms of IoT (Internet of Things), transmitting the data to the Blynk server and sending the relevant information about critical changes in the state of the battery to a mobile phone. Overvoltage, overcurrent, and overtemperature are used to protect the battery from damage in case these parameters exceed acceptable values; these protection features are to be realized with the help of the relay module to disconnect the load. Finally, there is an opportunity to estimate SoC.

2.2 MODULES OF THE PROJECT

The proposed system consists of various functional modules that have been designed, which perform distinct tasks related to battery monitoring, control, and protection systems. The modules are linked with one another through an STM32 microcontroller; and modules are linked through an ESP32 microcontroller as well.

2.2.1 Module 1 – Voltage Sensing Module (INA219)

In module 1, there is the usage of the INA219 sensor in order to sense the voltage of the battery directly. In this regard, the sensor can accurately sense the bus voltage without requiring the use of the voltage divider circuit. Through the I2C interface, communication between the sensor and the STM32 takes place.

2.2.2 Module 2 – Current Sensing Module (INA219)

INA219sensors use high accuracy current sensing and provide digital outputs via the I2C interface. This module facilitates the measurement of currents in charge-discharge operations.

2.2.3 Module 3 – Temperature Monitoring Module (LM35)

The LM35 temperature sensor produces an analog signal that is proportional to temperature readings. This is read and converted by the internal ADC of the STM32.

2.2.4 Module 4 – SoC Estimation Module

This module is responsible for calculating the SoC (State of Charge) of the battery, which shows its remaining capacity. The process is done based on the measurements of voltage and current readings in order to give the user an idea about the battery performance and the backup remaining. The calculation result will be shown on the LCD.

2.2.5 Module 5 – Display Module (LCD)

The display module displays various parameters related to the measurement like voltage, current, temperature, and SoC. This module offers a convenient interface to display battery information and status without any external device.

2.2.6 Module 6 – Load Simulation Module (Motor)

The load simulation module utilizes the system constraints for controlling the motor operation. This module is used to simulate the EV load that draws current from the battery.

2.2.7 Module 7 – Protection Module (Relay-Based Protection)

The function of the protection module is to safeguard the battery. It protects the battery during abnormal situations when there is an excessive value of voltage, current, and/or temperature, and disconnects the load automatically using the relay module.

2.2.8 Module 8 – IoT Communication Module (ESP32 + Blynk)

The ESP32 microcontroller facilitates wireless communication by transmitting real-time battery data to the Blynk IoT server. It also sends alerts in case of faults by using the mobile phone facility, thus making it possible to monitor and control remotely.

2.2.9 Module 9 – Central Processing Unit (STM32F407VG)

The STM32F407VG microcontroller serves as the brain of the proposed system. It receives data from all sensors, analyzes it, estimates the state-of-charge, triggers the relay to avoid faults, and communicates with peripheral devices. Its superior computing power and numerous interface choices qualify it for use in embedded systems such as BMS.

2.2 TASKS AND MILESTONES

Task / Milestone	Description	Status / Expected Output
Requirement Analysis	Study into EV battery management systems' necessities, standards, and research to determine the needs, constraints, and extent of the BMS project scope.	Completed – Defined project goals and system requirements
Hardware Selection	Selection of STM32F407VG Discovery board, INA219current sensor, LM35 temperature sensor, LCD, ESP32 module, relay module, and 3S battery pack.	Completed – Components finalized.
Circuit Design & Integration	Implementation of voltage sensing , sensor interfacing, relay protection circuit, and integration of all modules with STM32 microcontroller and ESP32 module.	Completed – Working prototype built.
Programming & Testing	Development of the embedded code for STM32 (ADC, I2C, LCD, safety circuitry, SOC computation) and ESP32 (communication within IoT using Blynk).	Completed – System testing and results
Protection Mechanism and Load Testing	Testing of overvoltage, overcurrent, and overtemperature conditions with relay-based cutoff system.Testing System using motor load	Completed- protection system and load operation verified
Data Monitoring & Alerts	Addition of ESP32 with Blynk platform to send real-time battery data and trigger mobile notifications during abnormal conditions.	Completed – Real-time alerts upon trigger on time
Final Implementation	Deploying of system for real-time Ev battery monitoring and protection	To Be Completed – Ready for demonstration.

Table 1 : Task Progress and Milestones

CHAPTER III

3. DESIGN OF BATTERY MANAGEMENT SYSTEM FOR EV APPLICATION

3.1 DESIGN APPROACH

- The proposed system is the intelligent Battery Management System (BMS) intended for application in electric vehicles (EV) with the STM32F407VG Discovery microcontroller used as the central processing unit of the system. The system has several sensing units: voltage and current sensing with the INA219 sensor, and temperature sensing with the LM35 sensor. The 16x2 LCD screen displays the values of the voltage, current, temperature, and State of Charge (SoC).
- Furthermore, the system that drives the motor load and provides realistic conditions for application in the EV environment. Also, an ESP32 microcontroller is used to provide an Internet-of-Things (IoT) capability, i.e., wirelessly transmit the battery parameter values to the Blynk online application on the mobile phone. Protection measures include a relay module to disconnect the load in case of overvoltage, overcurrent, and overtemperature.
- Data acquisition with sensors is constantly carried out, processed by the STM32 MCU, and SoC estimation occurs to update both local and remote interfaces with the parameter values. Real-time operation, safety, economy, and modularity of the solution make it appropriate for EV prototype-level application.

3.1.1 Codes and Standards

The design and development of this system adhere to industry standards in embedded systems and power electronics:

- **Embedded Hardware Standards:**Usage of regulated power supply units of 3.3V and 5V for the operation of STM32, ESP32, sensors, and peripheral circuits.

- **Communication Standards:**The I2C protocol is used for communications between the microcontroller and the INA219current sensor. INA219 is used for the analog measurement of voltages .UART/Serial is utilized for IoT communications between STM32 and ESP32.
- **Electrical Safety Standards**:Using the DC low-voltage system with relay isolation ensures safety from overvoltage, overcurrent, and overheating scenarios.
- **Coding Standards:**The programming language used for this project is embedded C (STM32CubeIDE) and Arduino IDE for ESP32. Modular code structures ensure efficiency in reading, modifying, and debugging programs.
- **Measurement Accuracy Standards**:Sensor calibrations are done in accordance with manufacturers' datasheets to provide correct voltages, currents, and temperatures.
- **System Reliability Standards:**Real-time monitoring of all components in real-time with preset threshold levels helps in achieving reliability and safety.
- **IoT and Data Transmission Standards:**Secure data transfer from sensors to the Blynk IoT platform.
- **Application Scope and Safety Note:**The system was designed only for monitoring and protection purposes of EV batteries and not meant as an industrial-grade BMS application.

3.1.2 Realistic Constraints

Several realistic limitations will influence the design of the proposed Battery Management System (BMS):

- **Power Limitation:**The system uses a constrained power supply source and must effectively manage energy usage. The choice of STM32, ESP32, sensors, and display units is made based on minimizing power usage while keeping other properties.

- **Cost Limitation:**It is intended to build an inexpensive proof-of-concept using inexpensive and easily accessible components such as INA219, LM35, and relay units.
- **Size Limitation:**The BMS design allows it to be easily incorporated into EV batteries, and its size makes it feasible.
- **Environment Limitation:**Sensors' readings may be influenced by environmental factors such as temperature changes, electrical noise, and changing loads, causing some errors in reading.
- **Accuracy Limitation:**SoC estimation and readings from sensors depend on calibration and measurement accuracy. Simple estimation algorithms used by the system do not guarantee high industrial standards.
- **Motor Load Limitation:**A motor load used for simulation may cause changing currents, affecting real-time readings and requiring robust filtering algorithms.
- **Internet Connectivity Limitation:**IoT data transmission using ESP32 requires a stable Wi-Fi connection, which may prevent timely data transfer and mobile notifications.
- **Scalability Limitation**:The design is meant for a 3S battery pack prototype and may need to undergo changes to accommodate large capacity or multi-cell EV batteries.
- **Safety Limitation**:While relay protection offers basic shutdown capabilities, it may not match the performance of modern solid-state safety devices used in industrial-grade BMS.

3.1.3 Alternatives and Trade-offs

During development, several alternatives were considered for sensors and components:

Component	Alternative Considered	Trade-off Decision
STM32F407VG Discovery	Arduino Mega / STM32F103	STM32F407VG chosen due to higher processing capabilities, multichannel ADC, improved peripheral capabilities, and real-time embedded application capabilities.
INA219 Current Sensor	ACS712	INA219 selected due to improved accuracy, I2C digital interface, and capability of measuring both current and voltage with high accuracy.
LM35 Temperature Sensor	DHT11 / DS18B20	LM35 selected due to linear analog signal output and improved accuracy in measuring battery temperatures.
Voltage Sensing	External Voltage Sensor Modules	INA219 Sensor: Provided stable voltage measurement of the 3S battery pack.
ESP32	Arduino Uno + Wi-Fi Module (ESP8266)	ESP32 selected due to inbuilt WiFi capability and improved processing capabilities.
LCD Display (16x2)	OLED Display	LCD due to low cost, convenience of connectivity, and adequate display capacity of parameters.

Relay Module	MOSFET-based Switching	Relay is used due to electrical isolation and ease of incorporating protection functions.
Motor Load	Resistive Load	Motor for simulation of realistic dynamic load conditions of an electric vehicle rather than a resistive load.
Blynk IoT Platform	ThingSpeak / Firebase	Blynk due to user-friendly interface, live monitoring, and seamless integration of mobile notifications.

Table 2: Alternative Considerations

These trade-offs ensured that the final design remained cost-effective, power-efficient, and user-friendly, without compromising functionality in the system for real time monitoring and protection

3.2 DESIGN SPECIFICATIONS

3.2.1 Hardware Design Table

Component	Function / Role	Interface Used	Power Supply	Remarks
STM32F407VG Microcontroller	The central processing unit; carries out the tasks of sensor data processing, SoC calculation, and the execution of protection algorithms.	GPIO, ADC, I²C, UART	3.3V	Core processing unit of the BMS.
INA219 Current Sensor	Measures battery current and power consumption.	I²C	3.3V–5V	High-accuracy digital current sensing.
LM35 Temperature Sensor	Measures battery temperature.	Analog Input (ADC)	5V	Provides linear output proportional to temperature.
LCD Display (16x2)	Displays voltage, current, temperature, and SoC values.	GPIO / I²C (if module used)	5V	Provides real-time user interface.
ESP32 Microcontroller	Handles IoT communication and sends data to the Blynk platform.	UART / GPIO	3.3V	Enables Wi-Fi connectivity and remote monitoring.
DC Motor (Load)	Acts as a load to simulate real-time battery usage.	DC motor	5V–12V	Represents EV load behavior.
Relay Module	Provides protection by disconnecting load during fault conditions.	GPIO	5V	Ensures safety during overvoltage, overcurrent, or overheating.

3S Lithium-ion Battery Pack	Main power source for the system.	-	~11.1V (nominal)	Represents EV battery system.
Power Supply Module	Regulates and distributes required voltages to components.	-	3.3V / 5V	Ensures stable operation of all modules.

Table 3: Design Specifications

3.2.2 System Explanation

- **Input Stage:** Sensors (INA219, LM35, and Voltage measuring INA219) monitor the current status of batteries.
- **Processing Stage:** STM32 evaluates sensor information to calculate the state of charge and check for abnormality conditions such as overvoltage, overcurrent, and overtemperature.
- **Output Stage:** The output is displayed by LCD and relay module that disconnects the load when an abnormal situation arises.
- **Communication Stage:** ESP32 transmits real-time data to Blynk IoT application to notify users through his mobile phone.

CHAPTER IV

4. PROJECT DEMONSTRATION

4.1 INTRODUCTION

- In continuation of our previous project of Smart Battery Management System (BMS) for electric vehicle applications, we now present the demonstration of a complete solution for real-time monitoring and protection of a 3S lithium-ion battery pack using embedded techniques. Our proposed system utilizes voltage and current sensing via INA219 sensor, and temperature sensing using the LM35 sensor, all under the control of STM32F407VG microcontroller.
- Furthermore, we have also included an interfacing of 16x2 LCD for displaying real-time parameters like voltage, current, temperature, and State of Charge (SoC). In addition to that, we have interfaced a load motor for creating realistic EV-like conditions. An ESP32 microcontroller has also been used in the proposed system for IoT-based monitoring, whereby the real-time data can be transferred to the Blynk platform in case of any abnormal conditions via mobile notifications. The overvoltage, overcurrent, and overtemperature conditions are detected by relays for protection purposes in the system.This demonstration showcases the working of individual modules and hardware components as well as real-time operation and results generated during test cases.

4.2 ANALYTICAL RESULTS

The obtained results have demonstrated effective performance and integration of all components of the BMS

- INA219 based sensing of voltage will help to accurately measure the voltage across the battery pack under different loads.

- The INA219current sensing circuit demonstrates correct operation while measuring currents both during charging and discharging processes and while the system is operated with load (motor).
- The LM35 temperature sensing mechanism correctly monitors the battery pack temperature depending on the conditions of use.
- The state of charge (SoC) estimation system offers reasonably accurate results about the battery pack capacity based on the measured data and displays them via an LCD screen.
- The system together with the DC motor successfully simulates conditions of usage when a device would have some kind of load attached.The relay-based protection mechanism quickly disconnects the load whenever there are faults such as overvoltage, overcurrent, or overtemperature issues.
- ESP32 transmits the collected data via Wi-Fi to the IoT platform (Blynk) where it can be accessed remotely from a smartphone and faults trigger mobile notifications.
- Thus, the presented system demonstrates reliable performance in monitoring the battery's parameters, SoC estimation, protection against possible faults, and Internet-of-things monitoring.

SIMULATION ANALYSIS:

The simulation for the proposed BMS has been carried out using the MATLAB/Simulink software in order to examine the SoC estimation and battery behavior when subjected to various load conditions. The design of the model uses the Coulomb counting technique, which involves integrating the battery current over time to estimate the state of charge left within the battery.

The State of Charge (SOC) is obtained by dividing the consumed charge by the total battery capacity to obtain a value that can be expressed in percentage terms. The initial state of charge is known to be used as the basis for full charge. In addition, there are signal processing elements in the block diagram that take care of the scaling of values.

<u>Simulation parameters:</u>

Parameter	Value
Battery Type	3S Lithium-ion
Battery Capacity	500 (mAh/Ah as per model)
Initial SoC	100%
Load	DC Motor
Simulation Time	100 seconds (or as used)

Table 4: Simulation constraints

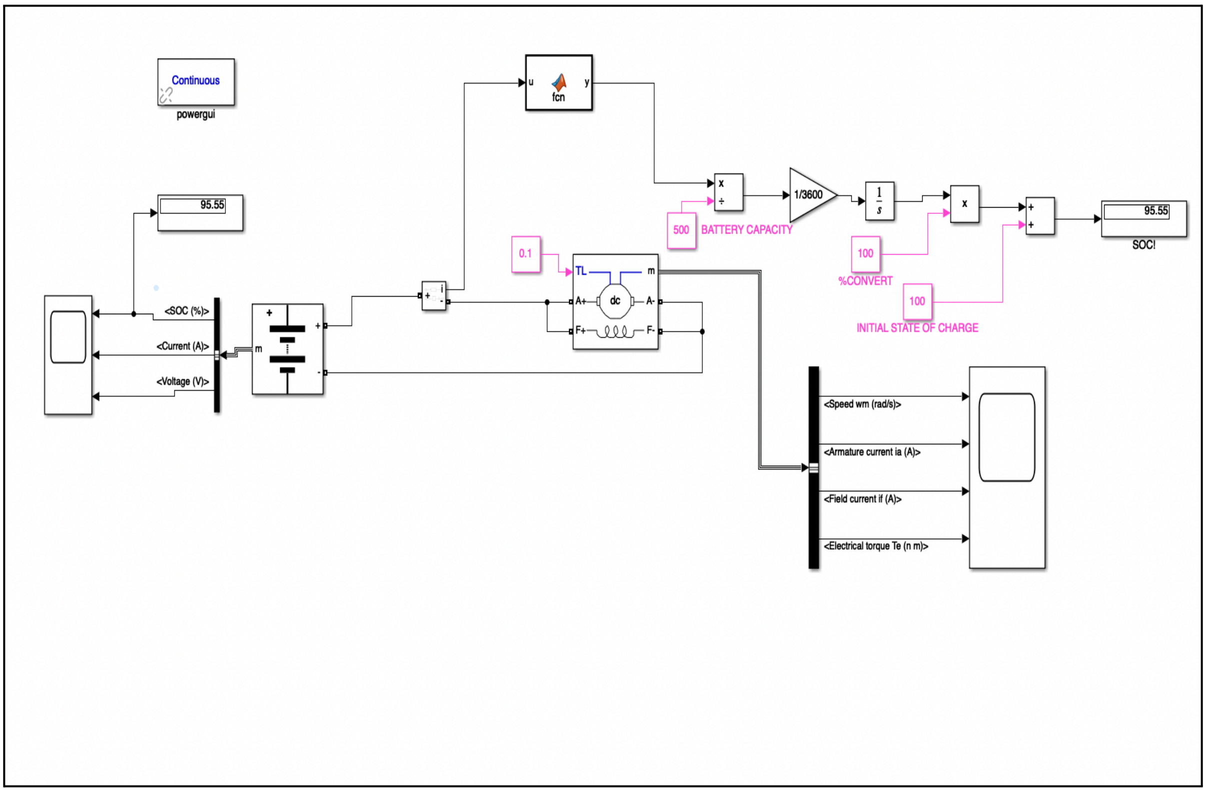

Figure 1.Simulation block diagram

Simulation results:

SoC vs Time:

- The State of Charge decreases gradually over time as the battery discharges under load.
- The graph shows a smooth decline, indicating proper integration of current and accurate SoC estimation.

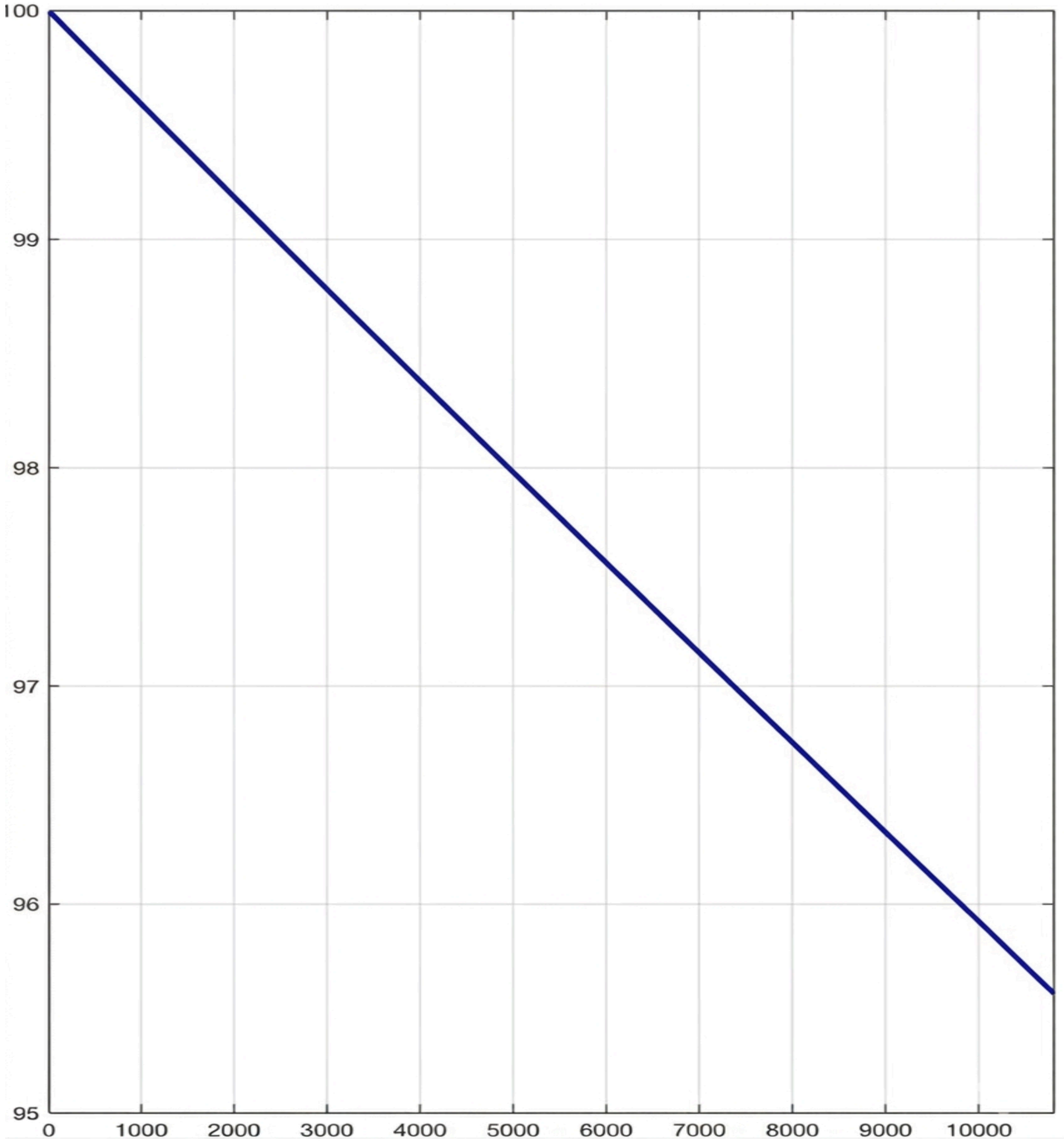

Figure 2 .SoC% vs Time(s)

<u>Current vs Time:</u>

- The current graph represents the load demand of the motor.
- Variations in current indicate dynamic load conditions, similar to real EV operation.

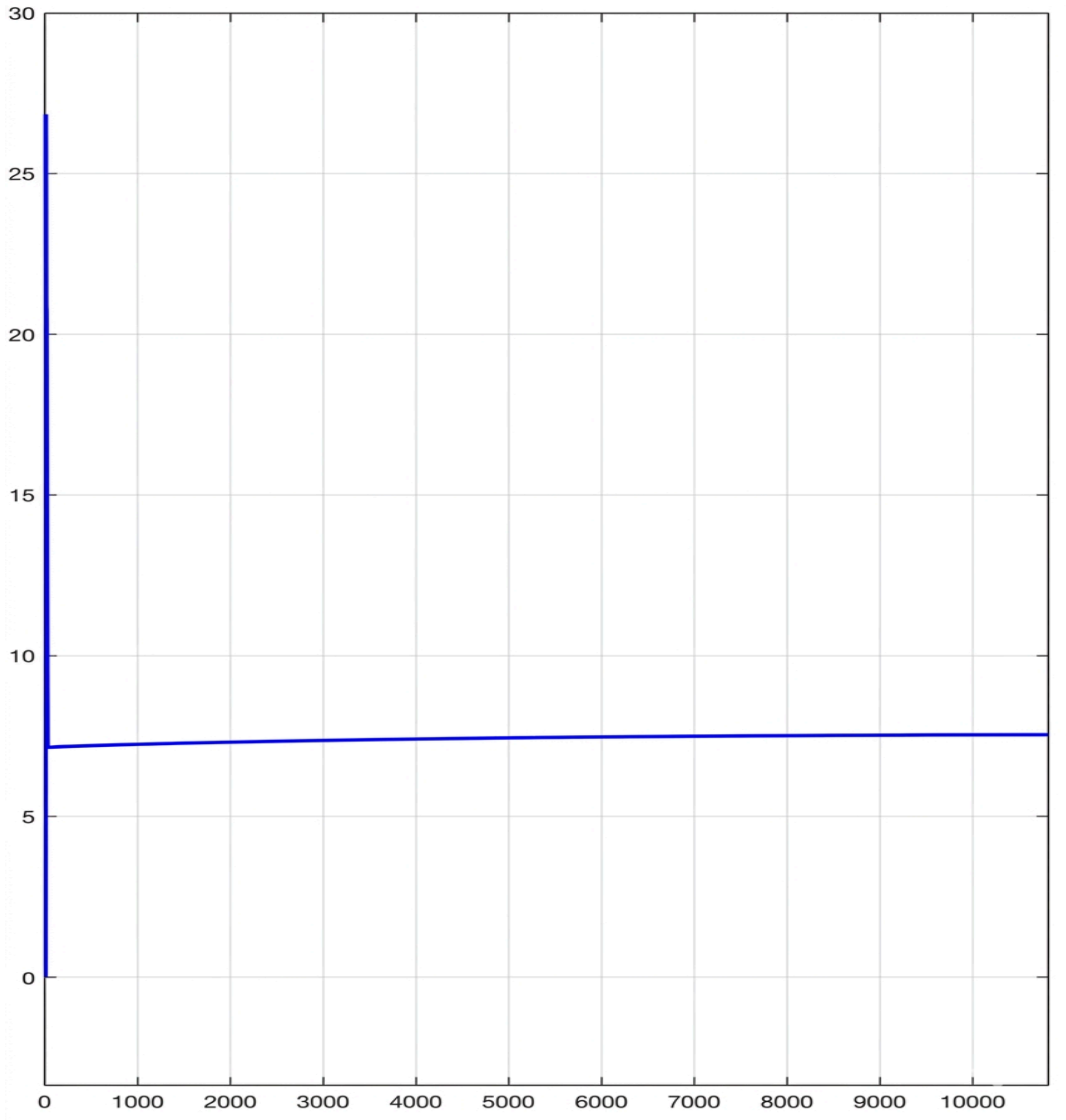

Figure 3 .Current(A) vs Time(s)

Voltage vs Time:

- The voltage graph shows a slight drop as the battery discharges thus reflecting realistic battery behavior under load.

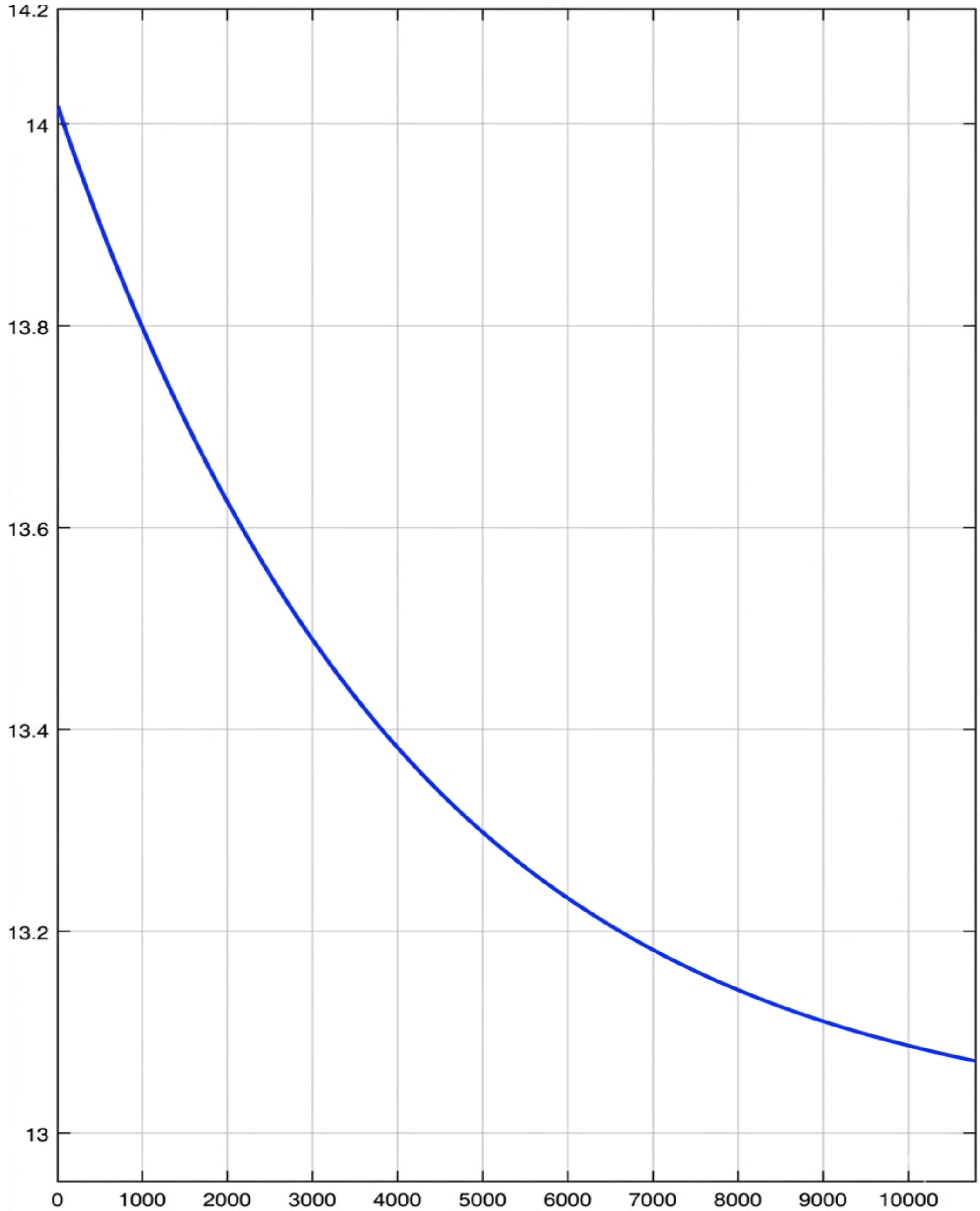

Figure 4 .Voltage(V) vs Time(s)

From the above results of the simulation, the SoC estimation technique using Coulomb counting proves to be successful. The SoC falls as a function of current utilization, which is a confirmation of the correctness of the mathematical model developed. The use of a motor load makes the current utilization vary, hence proving that the system can cater to dynamic situations.The mathematical model works perfectly well in tracking the discharge characteristics of the batteries, hence giving an estimate of the SoC.

4.3 HARDWARE RESULTS

The hardware was successfully developed and tested, integrating all modules using STM32F407VG Microcontroller and ESP32 for IoT communication.

<u>Observations:</u>

1. Voltage Measurement by INA219 Sensor:

- Provided stable voltage measurement of the 3S battery pack.
- The INA219 sensor was used to measure the battery voltage successfully, and the result was displayed accurately on the LCD screen.

2. INA219Current Sensor:

- Accurate measurement of charging and discharging currents in the system.
- Real-time current measurement accurate under varying loads of the motor.

3. LM35 Temperature Sensor:

- Proper sensing of the battery temperature.
- Linear output according to the temperature sensed by the device.

4. SoC Calculation:

- Accurate estimation of the State of Charge of the battery.
- The display showed appropriate values depending on the usage pattern of the battery.

5. LCD Display:

- Stable and continuous generation of outputs related to voltages, currents, temperatures, and state of charge of the battery.

6. DC Motor (Load):

- Smooth functioning of the motor with respect to real-time load conditions in the system.

7. Relay Module:

- Effective protection against overvoltage, overcurrent, and overheating of the system.
- Disconnecting the load automatically from the system in abnormal conditions.

8. ESP32 for IoT Communication:

- Successful uploading of real-time data to the Blynk Platform.
- Notifications sent via mobile phone during fault conditions.

9. STM32 Controller:

- Successfully analyzed all sensor inputs and performed appropriate actions.
- Operated consistently under consistent monitoring.

Conclusion:

All modules were in sync, and the performance was consistent, accurate, and real-time. The above analysis shows the effective implementation of sensors, processors, protection mechanisms, and communication via the Internet of Things, proving that our design of BMS is a good solution to be used in EVs.

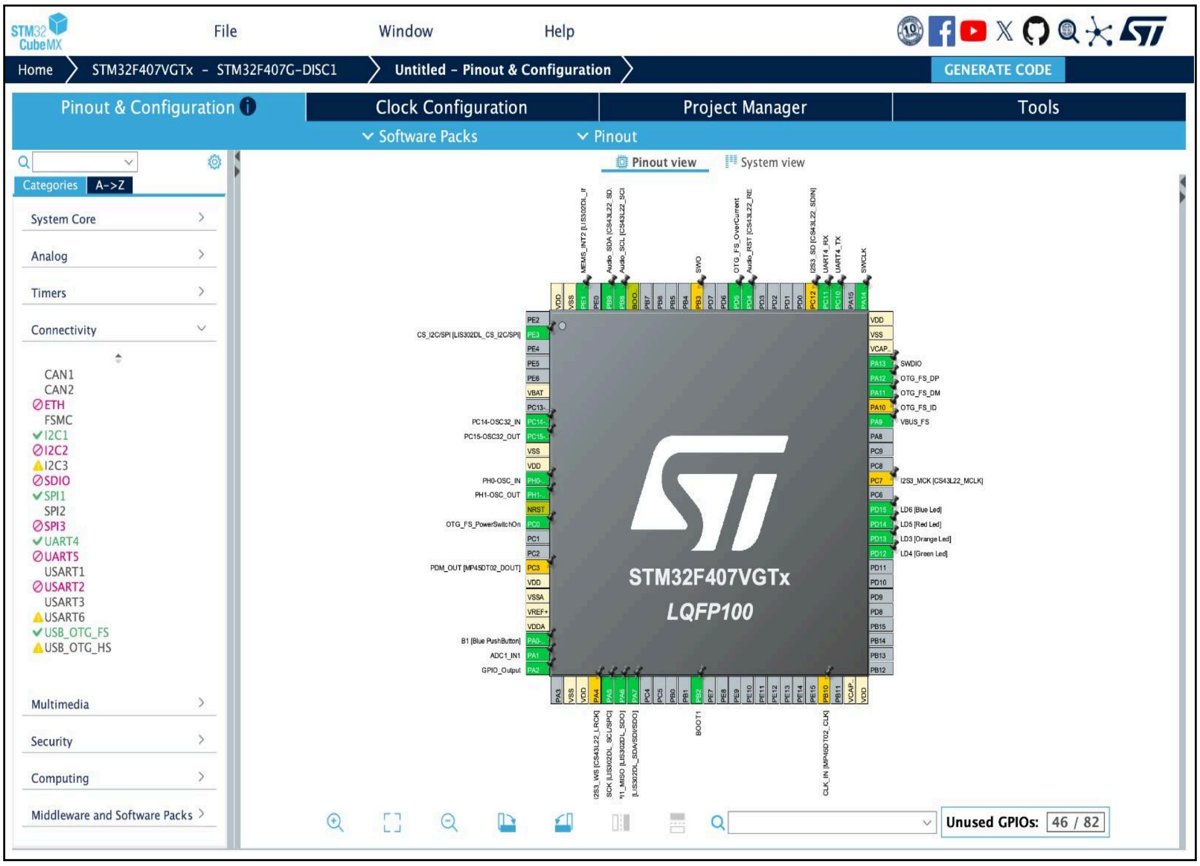

Figure 5. Setup Diagram of pins of STM32F407VG DISC 1 Board

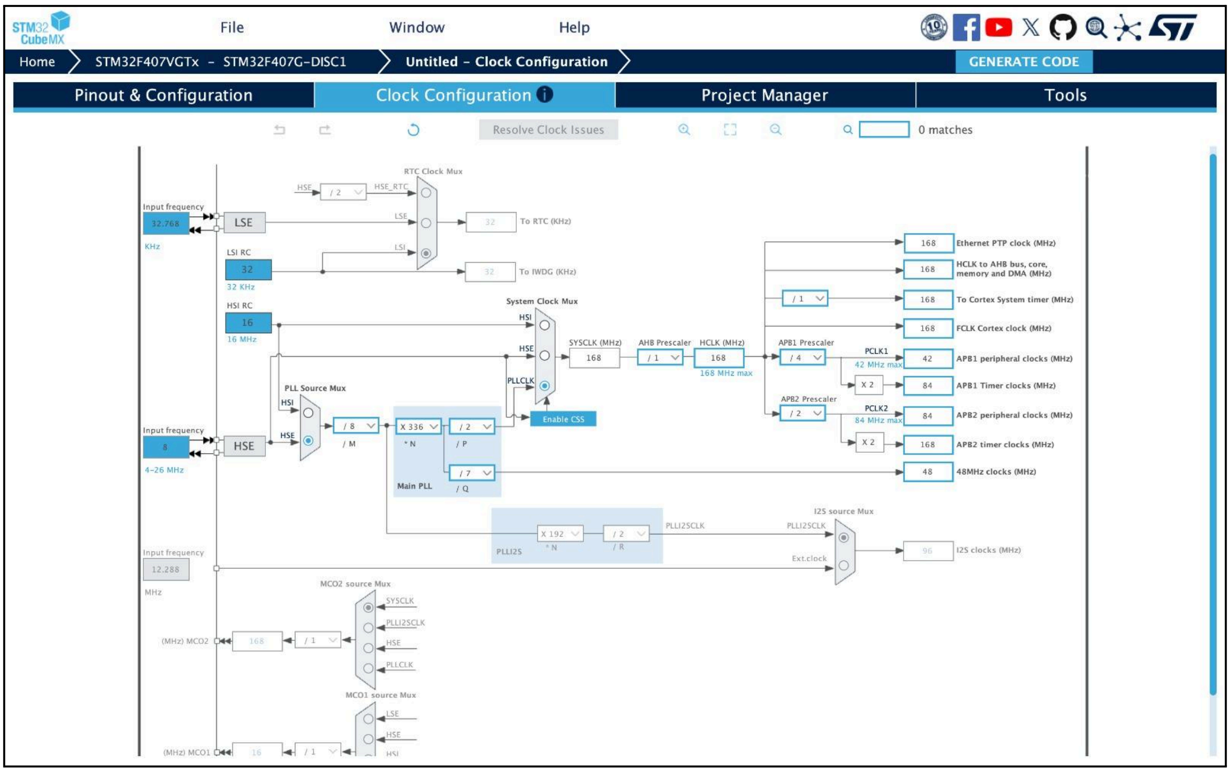

Figure 6. Clock configuration STM32F407VG DISC 1 BOARD

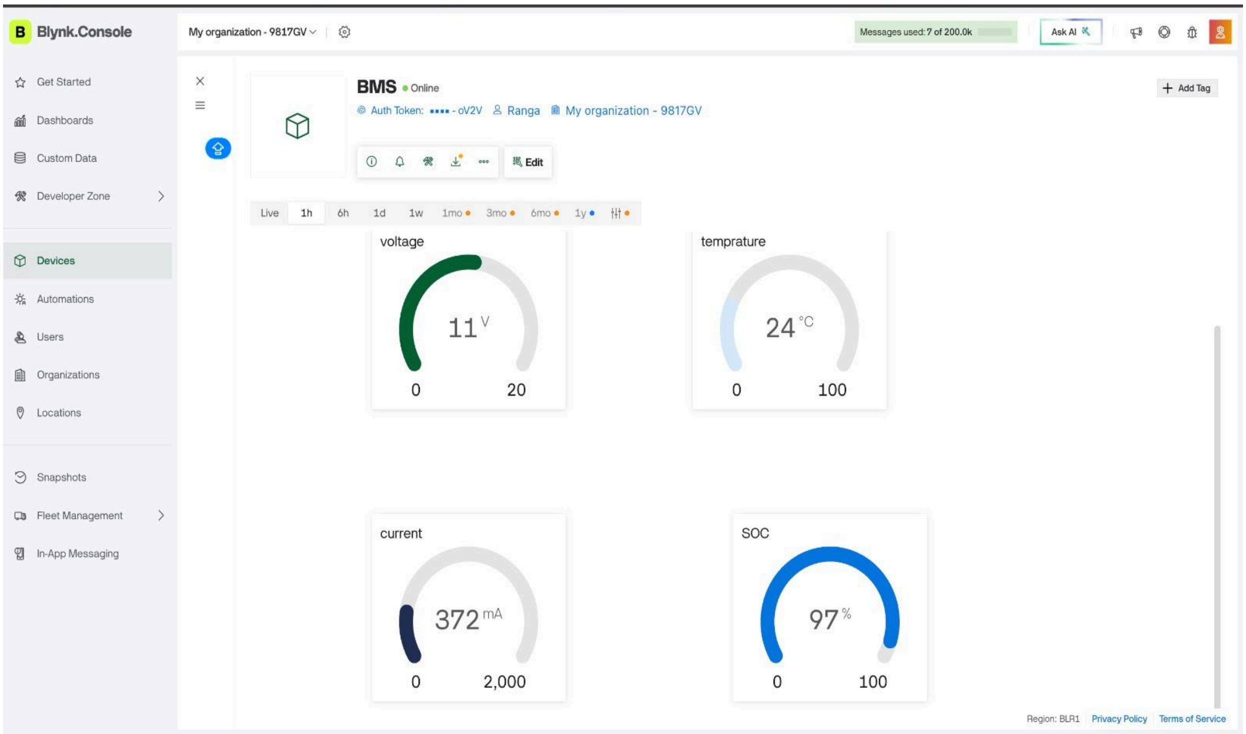

Figure 7.Readings of real time data from BLYNK WEBSITE

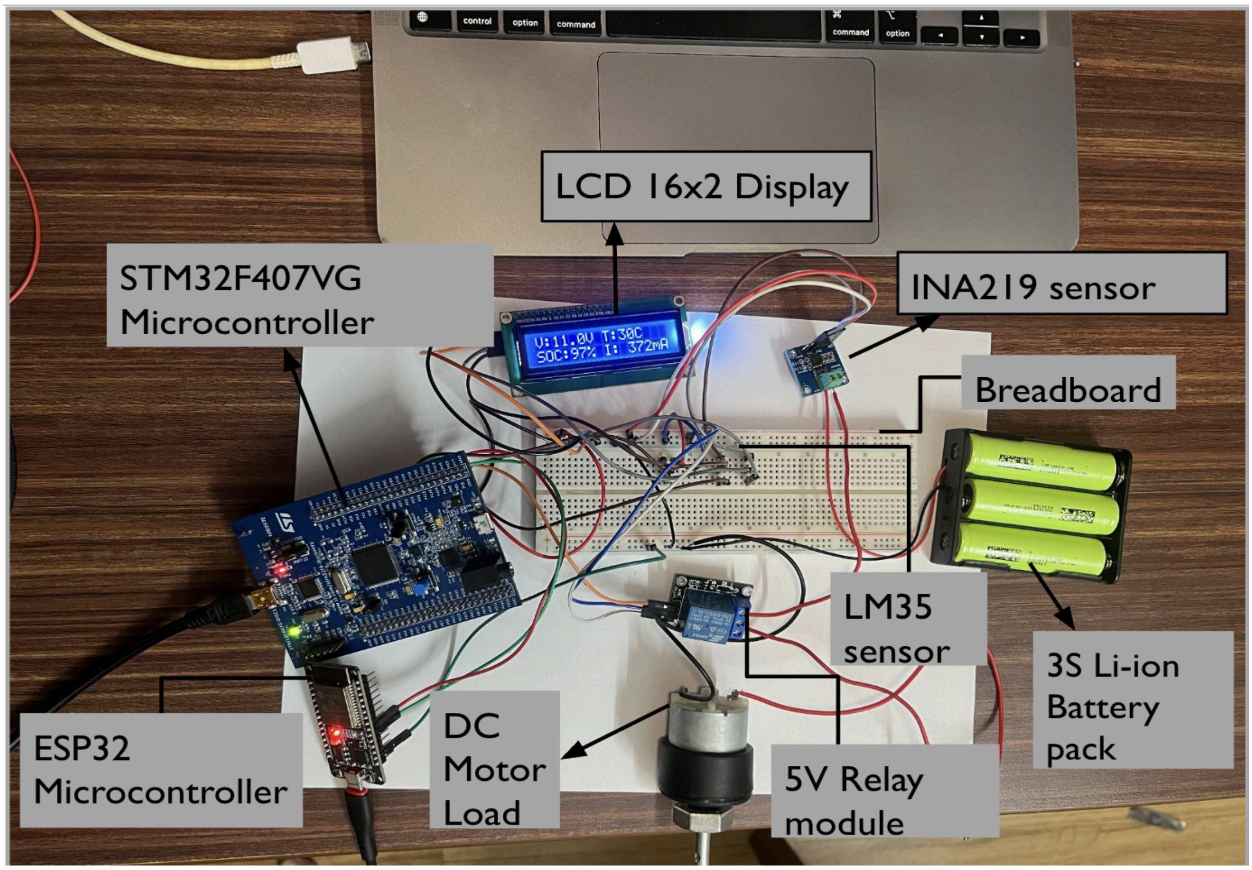

Figure 8.Hardware Connection Diagram

CHAPTER V

5. CONCLUSION

5.1 COST ANALYSIS

The proposed Smart BMS system for use in electric vehicles (EVs) is designed as an affordable prototype based on inexpensive components that are widely available. The entire development cost remains relatively low, thus making the design applicable for research purposes. The costs involved in this project are the microcontroller, sensor, IoT board, and hardware components.

<u>Cost Breakdown:</u>

Component	Quantity	Approx. Cost (INR)
STM32F407VG Discovery Board	1	₹2,200
INA219 Current Sensor	1	₹250
LM35 Temperature Sensor	1	₹120
ESP32 Microcontroller	1	₹450
16x2 LCD Display	1	₹200

DC Motor (Load)	1	₹300
Relay Module	1	₹150
3S Lithium-ion Battery Pack	1	₹900
Power Supply / Voltage Regulator	1	₹200
Connecting Wires, PCB, Misc. Components	-	₹300

Table 5: Cost analysis and approximation

Total Estimated Cost: ₹5,100 - ₹5,600

As shown above, the feasibility of the BMS system is not only economically viable but also offers important functions like real-time monitoring, protective features, SoC estimation, and IoT-based remote access.

5.2 SCOPE OF WORK

The scope of the proposed project is to develop an advanced Battery Management System (BMS) that can monitor the functioning of a lithium-ion battery pack used in electric vehicles in real-time.

Current Implemented Features:

- Voltage monitoring in real-time through STM32 microcontroller using INA219
- Current monitoring using INA219sensor.
- Temperature monitoring using LM35 sensor.
- State of Charge (SoC) estimation and its display through LCD.
- Display of battery parameters such as voltage, current, temperature, and state of charge in real-time.
- Simulation of load through a DC motor of 12V, 1000rpm
- Relay protection for overvoltage, overcurrent, and overheating of the battery.
- ESP32 integration for Internet of Things-based monitoring through Blynk.
- Mobile notifications in case of any irregularity in the battery's working condition.

Future Scope:

- Use of cell balancing techniques in case of a battery pack having multiple cells.
- Estimation and prediction of SoC and SoH values using advanced AI/ML algorithms.
- Long term storage of data into the cloud for analysis.
- Development of a dedicated mobile app to increase the usability of the system.
- Integration with a complete EV system including a battery pack.
- Implementation PCB design of the complete system for compact application and integration

5.3 SUMMARY

The project successfully manages to showcase a smart BMS design and implementation incorporating a variety of sensors and embedded systems. The key modules such as voltage detection, current detection, temperature detection, and SoC estimation modules operate in unison with the STM32F407VG microcontroller to analyze batteries in real-time accurately.

The safety aspect is provided via relay-protection schemes, while usability is assured via the use of an LCD monitor and IoT-based monitoring through ESP32 along with the Blynk platform. The capability to test the system under various realistic conditions due to the motor driver proves the practicality of the designed system.

In conclusion, it can be stated that the project has managed to achieve the goal of designing a cost-effective, safe, and user-friendly BMS. In addition, it proves how much potential there is in the integration of the field of embedded systems with IoT technology. Future development may involve AI-based analysis for advanced battery diagnostics.

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